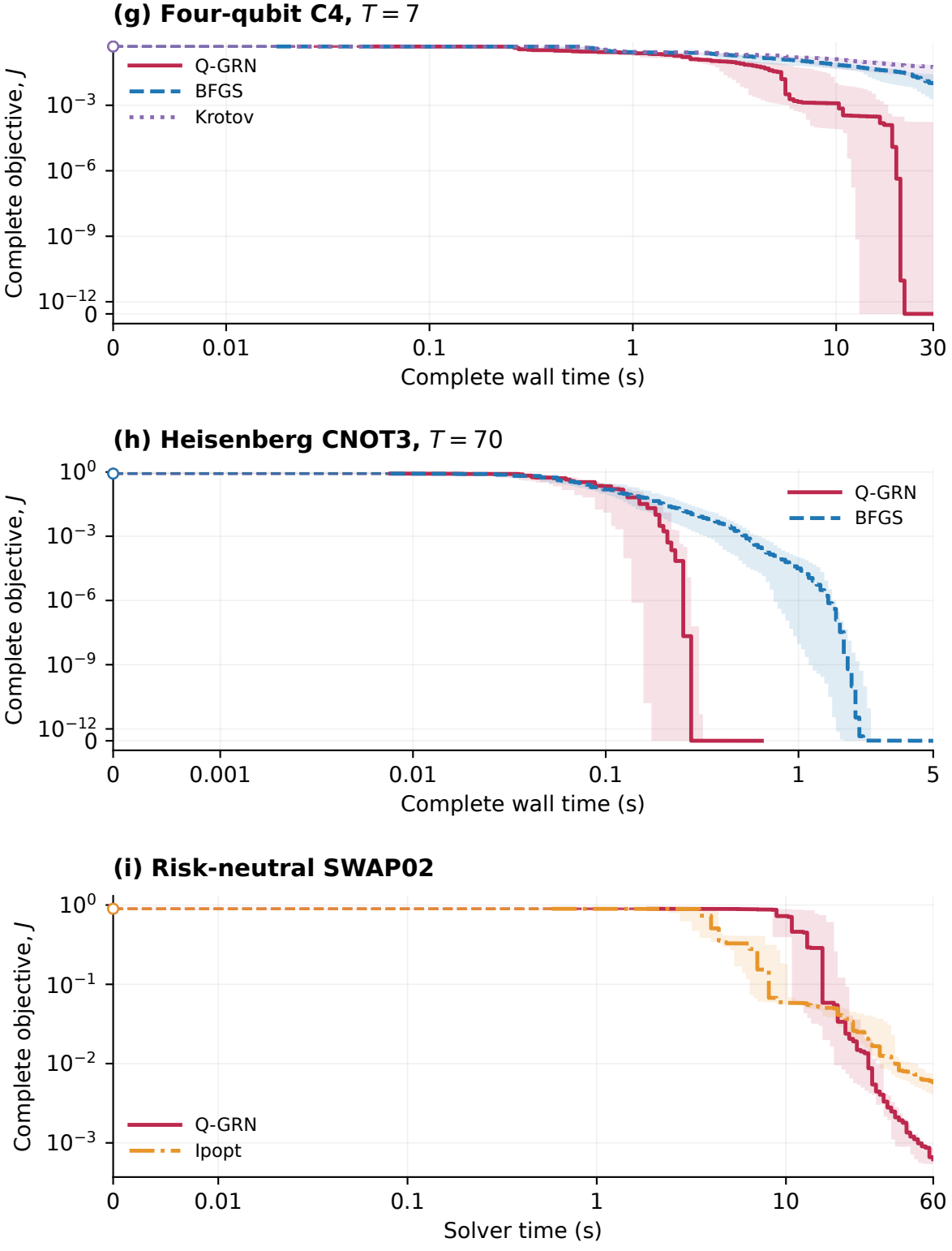
(g) Four-qubit C4, $T = 7$
Q-GRN
BFGS
Krotov
Complete objective, J
10^{-3}
10^{-6}
10^{-9}
10^{-12}
0
0
0.01
0.1
1
10
30
Complete wall time (s)
(h) Heisenberg CNOT3, $T = 70$
Q-GRN
BFGS
Complete objective, J
10^{0}
10^{-3}
10^{-6}
10^{-9}
10^{-12}
0
0
0.001
0.01
0.1
1
5
Complete wall time (s)
(i) Risk-neutral SWAP02
Q-GRN
Ipopt
Complete objective, J
10^{0}
10^{-1}
10^{-2}
10^{-3}
0
0.01
0.1
1
10
60
Solver time (s)